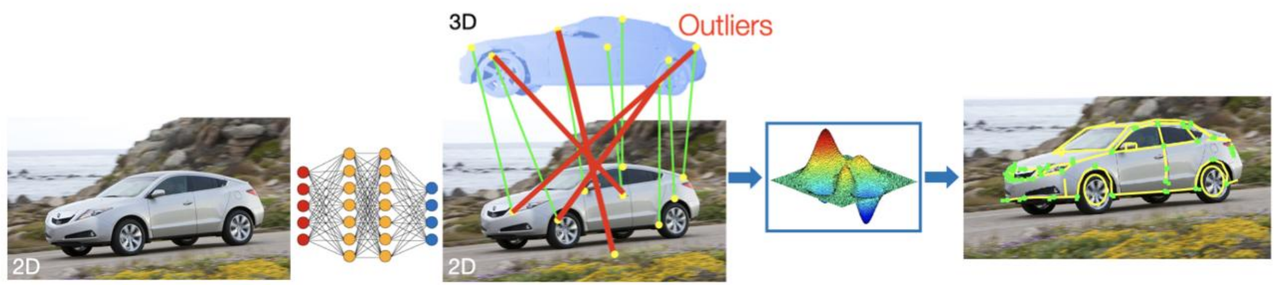
Conformal Semantic Keypoint Detection with Statistical Guarantees

Heng Yang and Marco Pavone

Workshop on Robot Learning: Trustworthy Robotics

Object Pose Estimation: why untrustworthy?



Stage 1: keypoint detection

Stage 2: pose estimation (optimization)

Keypoints are not guaranteed to be close to groundtruth

Focus of our paper

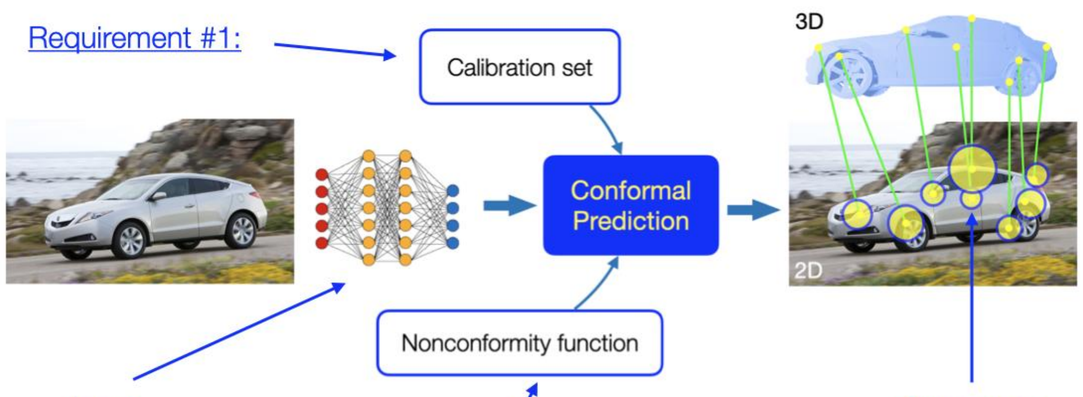
Popular heuristics (e.g., RANSAC) get trapped in local minima and fail without notice

Related work: certifiable algorithms [YC22TPAMI]

Our method: conformal keypoint detection

Paradigm shift: from heuristic prediction to probabilistically correct prediction sets

Key ingredient: inductive conformal prediction (ICP) — finite-sample uncertainty quantification without assumptions on the data distribution or the prediction model



Input: Heuristic prediction (e.g., heatmap of the keypoint location)

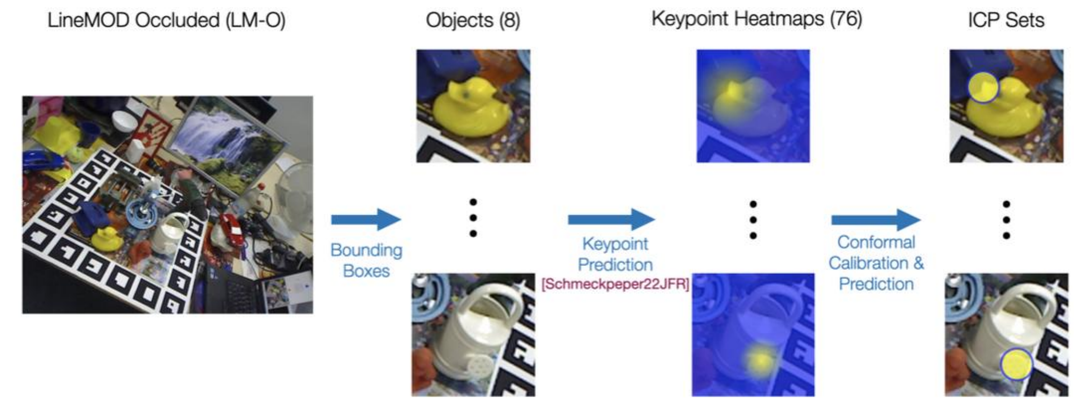
Requirement #2: See paper for details!

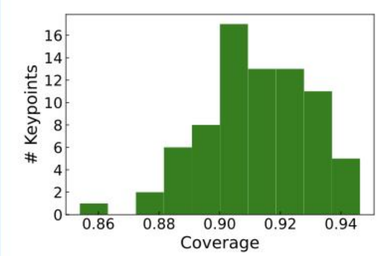
Guarantee: Cover the true keypoints with a user-specified probability (e.g., 90%)

Experiments: works out of the box!

Setup: 200 images for calibration, 1214 images for testing

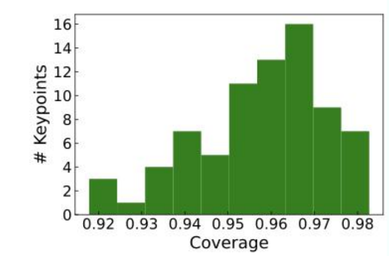
User: give me 90% and 95% coverage of the true keypoints

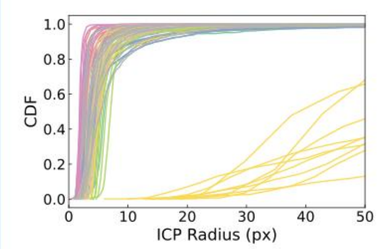


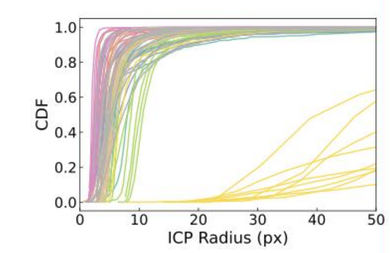
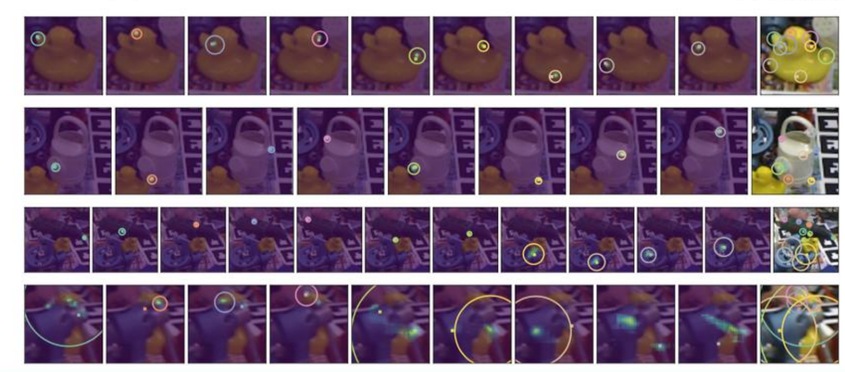


Valid coverage

←90%, 95%→





Small size

←90%, 95%→